THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

Dec 13, 2021 9:30 a.m. – 12:00 noon

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL SIX questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (24 points) Prove or disprove each of the following statements:

   (a) There exist sets A and B such that $A \neq B$ and $\mathcal{P}(A) = \mathcal{P}(B)$

   (b) The only prime p which satisfies $5p + 9 = n^2$ for some $n \in \mathbb{N}$ is $p = 11$.

   (c) Let R be the relation on $\mathbb{N}$ defined by xRy if $xy = n^2$ for some $n \in \mathbb{N}$. Then R is an equivalence relation on $\mathbb{N}$.

   (d) For any nonempty infinite sets A and B, if A is uncountable and there is a surjective function from A to B, then B is uncountable.

2. (14 points)

   (a) Prove that there exist infinitely many positive real numbers a such that $a + \frac{1}{a}$ is an integer.

   (b) Let a be any positive real number such that $a + \frac{1}{a}$ is an integer. Use the strong form of induction to prove that

$$a^n + \frac{1}{a^n}$$

   is also an integer for any positive integer n.

3. (20 points)

   (a) Prove that for any nonnegative integer m,

$$5^m \equiv \begin{cases} 1 \pmod 8 & \text{if } m \text{ is even,} \\ 5 \pmod 8 & \text{if } m \text{ is odd.} \end{cases}$$

   (b) Prove that for any nonnegative integers x, y, if each of x and y can be expressed as the sum of two perfect squares, then so can the product xy. Hence, or otherwise, prove that 5^m can be expressed as the sum of two perfect squares for any nonnegative integer m.

   (c) For any nonnegative integers m, n, give a necessary and sufficient condition for $5^m + 5^n$ being able to be expressed as the sum of two perfect squares. Prove that your condition is indeed necessary and sufficient. (*Hint*: Use the results in (a) and (b).)

4. (16 points)

(a) Let $f : \mathbb{N} \times \mathbb{N} \longrightarrow \mathbb{N}$ be a function defined by

$$f(m,n) \;=\; 2^{m-1}(2n-1).$$

Prove that f is bijective.

(b) For each $k \in \mathbb{N}$, let $S_k = \{x \in \mathbb{N} : x \equiv 2^{k-1} (\text{mod } 2^k)\}$. Prove that $P = \{S_k : k \in \mathbb{N}\}$ gives a partition of $\mathbb{N}$.

5. (14 points) Let A and B be nonempty sets, and let $f : A \longrightarrow B$ be a function. For any nonempty proper subset C of A, define the *restriction* $f|_C$ of f to C to be the function $f|_C : C \longrightarrow B$ given by $(f|_C)(c) = f(c)$ for all $c \in C$. <u>Prove or disprove</u> each of the following:

(a) $(f|_C)(X \cap C) = f(X) \cap (f|_C)(C)$ for any $X \subseteq A$.

(b) $(f|_C)^{-1}(Y) = f^{-1}(Y)$ for any $Y \subseteq B$.

(c) $(f|_C)^{-1}(B \setminus Y) = C \setminus f^{-1}(Y)$ for any $Y \subseteq B$.

6. (12 points) Let

$$\begin{aligned} S &= \{x \in \mathbb{R} : x = m\sqrt{2} + n\sqrt{\pi} \text{ for some } m, n \in \mathbb{Z}\} \text{ and} \\ T &= \{x \in \mathbb{R} : x = p\sqrt{2} + q\sqrt{\pi} \text{ for some prime numbers } p, q\} \end{aligned}$$

(a) Prove that S is denumerable.

(b) <u>Prove or disprove</u>: There exists a bijective function from S to T.

(*Hint*: You may use without proof the fact that $\sqrt{2}$ and π are irrational.)

******* **END OF PAPER** *******